LAB REPORT: AWS VPC PEERING CONFIGURATION

Course: Cloud Infrastructure Lab | **Department:** Computer Science & Technology
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1. Executive Summary & Learning Objectives

This lab report documents the successful architecture, implementation, and traffic validation of an isolated Virtual Private Cloud (VPC) Peering connection within Amazon Web Services (AWS). In cloud environments, production architectures require distinct, isolated network domains to separate concerns (such as staging vs. production, or shared services vs. core applications). By default, resources in separate VPCs cannot communicate via private IPv4 channels.

The core learning objectives achieved during this exercise include:

- **Network Segregation:** Provisioning distinct VPC topologies with non-overlapping Classless Inter-Domain Routing (CIDR) blocks to ensure absolute data tier separation.
- **Inter-VPC Transit Routing:** Building, authorizing, and establishing an active VPC Peering lifecycle handshake across disjointed infrastructure components.
- **Asymmetric Route Table Mutation:** Explicitly configuring localized CIDR target paths to forward cross-network frames directly through the peer bridge interface rather than routing packets into the public internet.

2. Network Provisioning Asset Matrix

To ensure deterministic pathing without IP allocation collisions, two independent VPC spaces were generated. The complete asset parameters are summarized in the table below:

Asset Identifier	IPv4 CIDR Block	Subnet Mapping	Primary Component Function
VPC A (Source Space)	10.0.0.0/16	Subnet-A: 10.0.1.0/24	Hosts primary transactional and application compute nodes demanding secure outbound integration pipelines.
VPC B (Target Space)	192.168.0.0/16	Subnet-B: 192.168.1.0/24	Hosts isolated data tier systems, analytical processing repositories, and central backup environments.

pcx-bridge (Peering Link)	N/A (Cross-VPC Logical Router)	Binds Subnet-A and Subnet-B	Establishes full-duplex, low-latency private routing between peered clusters using AWS internal fabrics.

3. Operational Implementation Workflow

The practical deployment followed a rigid, high-fidelity infrastructure layout across three primary phases:

Phase 1: Network Space Setup
Both VPC structures were spun up using standard configurations with DNS Hostnames and Resolution enabled. An EC2 instance was deployed inside each public/private subnet boundary to act as operational testing vectors.

Phase 2: Peering Handshake Authorization
A peering request was initialized from VPC A targeting VPC B. Because both environments exist under standard administrative scopes, the connection state moved from *Initiating-Request* to *Pending-Acceptance*, where it was manually accepted to activate full-duplex tunnel linkage.

Phase 3: Route Table Mutation
To guarantee packets knew how to locate the adjacent network, localized route tables were updated explicitly:

- **Route Table A (VPC A):** Appended a static path mapping destination 192.168.0.0/16 target pcx-bridge.
- **Route Table B (VPC B):** Appended a static path mapping destination 10.0.0.0/16 target pcx-bridge.

4. Connectivity Verification & Analytics Matrix

To ensure proper cross-network routing and isolate variables, diagnostic ICMP packages were transmitted from the compute node inside VPC A to the target node inside VPC B.

Test ID	Source Node	Destination Node	Protocol / Metrics	Execution Outcome Status
TC-01	Compute-A (10.0.1.15)	Compute-B (192.168.1.42)	ICMP Ping Echo Request Count: 5 packets	**SUCCESSFUL** 0% Packet Loss Avg latency: 1.8ms
TC-02	Compute-A (10.0.1.15)	Compute-B (192.168.1.42)	TCP Custom Transport Layer Port: 8080 Telnet Bound	**SUCCESSFUL** Connection handshake established

TC-03	Compute-B (192.168.1.42)	Compute-A (10.0.1.15)	Reverse Path Echo Check Count: 5 packets	**SUCCESSFUL** Asymmetric routing confirmed active

5. Conclusion & Architecture Integrity Verification

The successful cross-transmission metrics confirm that the network architecture functions perfectly. Both VPC environments route internal data securely within the AWS backbone mesh network, maintaining robust isolation bounds while eliminating extra hops and public internet exposure.